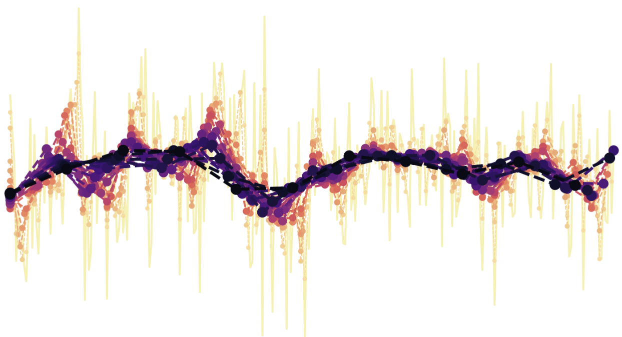
Ab initio Nuclear Theory: A Crash Course

Speaker: *Georgios Palkanoglou (TRIUMF)*
Dates: 1–5 June, 10-11am [11am–1pm on 5 June, computational lab]
Location: Aula Pere Pascual, Facultat de Física and online:
https://ub-edu.zoom.us/j/91310900080?pwd=ZKktP8bImwFRgXg4TRBsks0yrnsLqw.1

Overview

*This **intensive course** will provide an introduction to **modern ab initio nuclear theory**, focusing on fundamental concepts and state-of-the-art computational approaches to the nuclear many-body problem. The course is designed for graduate students & postdoctoral researchers interested in computational methods, nuclear structure and quantum many-body physics.*



Contents

1. Introduction to one-body, two-body, and many-body correlations
2. Pairing correlations: going beyond the Slater determinant(s)
3. Nuclear forces (and how to fake them)
4. *Ab initio* approaches to the nuclear many-body problem: Quantum Monte Carlo
5. *Ab initio* approaches to the nuclear many-body problem: No-Core Shell Model
6. **Computational Lab:** solving the BCS gap equations for neutrons (using Chiral EFT forces) and for cold atomic systems

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Organizers / Support

This course is an initiative funded by the **Catalonia Quantum Academy** and organized by the **Hadron and Nuclear Physics** group at the **ICCUB** (Institut de Ciències del Cosmos, UB). For any questions, please contact: arnau.rios@ub.edu